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22TP302 QUANTITATIVE APTITUDE AND LOGICAL REASONING

Hours Per Week :

L	T	P	C
0	4	0	2

PREREQUISITE KNOWLEDGE: Basic logical thinking and Problem solving ability.

COURSE DESCRIPTION AND OBJECTIVES:

The Students will be introduced to various Arithmetic and Reasoning Problems. The students will have acquaintance with various problems like Time & Work, Time & distance, Percentages, Profit & Loss etc. besides solving puzzles and Critical Reasoning.

MODULE-1

UNIT-1

0L+4T+12P=16 Hours

Number system, LCM & HCF of numbers, Percentage, Ratio and proportion, Profit, loss and discount, Average & Mixtures, Simple Interest & Compound interest.

UNIT-2

0L+4T+12P=16 Hours

Time and work, Time & distance, Problems on trains, Problems on ages, Permutation & Combinations, Probability.

PRACTICES:

- Each concept would be taught in detail in the class followed by 10 problems solved in the class.
- Students would have to solve 10 additional problems as a homework assignment in each concept.

MODULE-2

UNIT-1

0L+4T+12P=16 Hours

Number series, Letter series, Analogy, Odd man out, Coding and decoding, Syllogisms- Statement & Conclusions, Puzzle test.

UNIT-2

0L+4T+12P=16 Hours

Blood relations, Direction sense test, Order & Ranking, Seating Arrangements, Calendar & Clocks.

PRACTICES:

- Each concept would be taught in detail in the class followed by 10 problems solved in the class. Students would have to solve 10 additional problems as homework assignments in each concept.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Meet the demands of current job market besides equipping them higher studies like CAT, GMAT etc.	Apply	1	2, 5
2	Solve Arithmetic and Reasoning Problems within shortest possible time without paper work.	Apply	1	2, 5
3	Exhibit better analytical skills and aptitude skills.	Analyse	2	2, 4
4	Develop interpretational skills.	Evaluation	2	2, 4

SKILLS:

- ✓ *Helps in developing and improving problem-solving skills.*
- ✓ *Flexing and honing logical abilities.*
- ✓ *Allow students to develop critical thinking skills.*

TEXT BOOKS:

1. R. S. Aggarwal, "Quantitative Aptitude for Competitive Examinations", S. CHAND Publications- Revised Edition, 2017.
2. ARIHANT, "A New Approach to Verbal & Non-Verbal Reasoning", Arihant Publication- Revised Edition, 2021.

REFERENCE BOOKS:

1. Trishna Knowledge Systems, "Quantitative Aptitude for Competitive Examinations", Pearson Publication, 2013.
2. R. S. Aggarwal, "A Modern Approach to Verbal & Non-Verbal Reasoning", Revised Edition, S. CHAND Publications, 2018.

22CS401 CRYPTOGRAPHY AND NETWORK SECURITY

Hours Per Week :

L	T	P	C
2	0	2	3

PREREQUISITE KNOWLEDGE: Computer networks.

COURSE DESCRIPTION AND OBJECTIVES:

This course focuses on the modern concepts of network security using various cryptographic algorithms and underlying network security applications. It enables to understand various symmetric and asymmetric cryptographic techniques. It focuses on providing security services in applications such as e-mail functioning, web security and secure electronic transactions protocol and system security.

MODULE-1

UNIT-1

12L+0T+8P=20 Hours

INTRODUCTION

Introduction To Computer and Network Security Concepts: Computer Security Concepts, Security attacks, Security services, Security mechanisms, Fundamental Security Design Principles, Attack Surfaces and Attack trees, A model for network security.

Classical Encryption Techniques: Symmetric cipher model, Substitution techniques, Transposition techniques

UNIT-2

12L+0T+8P=20 Hours

SYMMETRIC AND ASYMMETRIC CRYPTOGRAPHY

Symmetric Ciphers: Block cipher principles, Data encryption standard, Strength of DES, Blockcipher design principles, AES cipher, Multiple encryption and triple DES, Block cipher modes of operation, RC4.

Asymmetric Ciphers and Cryptographic Hash Functions: Principles of public key cryptosystems, RSA algorithm, Diffie-Hellman Key Exchange, Message Authentication requirements, Authentication functions, Message authentication Codes, Hash functions, Security of hash functions and MACs, Digital signature standard.

PRACTICES:

- Implement Substitution and Transposition Ciphers
 - Ceaser cipher
 - Playfair cipher
 - Hill cipher
 - Rail fence cipher
- Implement Symmetric Cipher
 - S-DES
 - RC4
- Implement Asymmetric Cipher
 - RSA
 - Diffie-Hellman
 - Hash Function

Source: https://www.brainkart.com/subject/CRYPTOGRAPHY-AND-NETWORK-SECURITY-PRINCIPLES-AND-PRACTICE_136/

MODULE-2**UNIT-1****12L+0T+8P=20 Hours****SECURITY APPLICATIONS****Network Security Applications:** Kerberos, X.509 authentication service, Public key infrastructure,**E-Mail Security:** Pretty good privacy, S/MIME.**IP Security Overview:** IP security architecture, Authentication header, Encapsulating security payload, Combining security associations, key management.**UNIT-2****12L+0T+8P=20 Hours****WEB AND SYSTEM SECURITY****Web Security:** Secure socket layer and transport layer security, HTTPS, Secure Shell (SSH)**System Security:** Intruders, Intrusion detection, Malicious software, Firewalls**PRACTICES:**

- Configure IP Address in a system in LAN (TCP/IP Configuration)
- Configure DNS to establish interconnection between systems
- Configuring Windows Firewall
- Adding users, setting permissions
- Configure Mail server
- Demonstrate the usage of Wireshark to identify abnormal activity in network communication.
- Demonstrate usage of NMAP (Zenmap) Tool in Network Scanning.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Apply cryptographic techniques in various security service solutions effectively in everyday professional and social contexts.	Apply	1,2	1,2
2	Analyze the usage of secure protocols to safeguard sensitive data using internet.	Analyze	1,2	1,2
3	Usage of tools to Identify abnormal activity in network communication to take appropriate action.	Apply	2	5
4	Apply various security protocols to safe guard the data internet using SSL/TCL.	Apply	2	1,2

TEXT BOOK:

1. William Stallings, "Cryptography and Network security", 7th Edition, Pearson Education, 2017.

REFERENCE BOOKS:

1. William Stallings "Network Security Essentials Applications and Standards", 2nd Edition, Pearson Education, 2009.
2. Eric Malwald, "Fundamentals of Network Security", 4th Edition, Pearson Education, 2010.
3. Buchmann, "Introduction to Cryptography", 2nd Edition, Pearson Education, 2009.
4. Charlie Kaufman, "Radis Perlman and Mike Speciner, Network Security - Private Communication in a Public World", 1st Edition, Pearson Education, 2009.

SKILLS:

- ✓ Design various security services for appropriate applications.
- ✓ Identifying the appropriate firewall, password management and anti-virus models for specific applications.
- ✓ Test and resolve threats and malfunctions in network.
- ✓ Apply different security mechanisms for web applications.
- ✓ Build authentication system for security protocols.

22CS311 PARALLEL AND DISTRIBUTED COMPUTING

Hours Per Week :

L	T	P	C
2	2	0	3

PREREQUISITE KNOWLEDGE: Fundamental of programming and operating system, computer networking, and computer Organization..

COURSE DESCRIPTION AND OBJECTIVES:

This course covers the challenges faced in constructing parallel and distributed applications, Various implementation techniques, paradigms, architectures and parallel algorithms. And current trends in parallel and distributed computing like Open MP, POSIX Threads, Apache Hadoop (DFS).

MODULE-1

UNIT-1

12L+8T+0P=20 Hours

INTRODUCTION TO PARALLEL COMPUTING

The idea of Parallelism, Power and potential of parallelism, examining sequential and parallel programs, Scope and issues of parallel and distributed computing, Goals of parallelism, Parallelism and concurrency using multiple instructions streams.

Parallel Architecture: Pipeline architecture, Array processor, Multi-processor architecture, Systolic architecture, Dataflow architecture, Architectural classification schemes, Memory access classification, Memory Issues: Shared vs. distributed, Symmetric multiprocessing (SMP), SIMD, Vector processing, GPU co-processing, Flynn's Taxonomy, Instruction Level support for parallel programming, Multiprocessor caches and Cache Coherence, Non-Uniform Memory Access (NUMA).

UNIT-2

12L+8T+0P=20 Hours

PARALLEL ALGORITHM DESIGN PRINCIPLES AND PROGRAMMING

Need for communication and coordination/synchronization, Scheduling and contention, Independence and partitioning, Task- Based Decomposition, Data Parallel Decomposition, Characteristics of task and interaction, Load balancing, Data Management, parallel algorithm models, Sources of overhead in parallel programs, Performance metrics for parallel algorithm implementations, Parallel algorithmic patterns like divide and conquer, Map and Reduce, Specific algorithms like parallel Merge Sort, Parallel graph Algorithms.

PRACTICES:

- Identify Multiple Instruction Single Data, or MISD. How would an MISD system work? Give an example.
- Suppose a shared-memory system uses snooping cache coherence and write-back caches. Also suppose that core 0 has the variable x in its cache, and it executes the assignment $x = 5$. Finally suppose that core 1 doesn't have x in its cache, and after core 0's update to x, core 1 tries to execute $y = x$. What value will be assigned to y? Why?
- Consider a simplified version of bucket-sort. You are given an array A of n random integers in the range $[1..r]$ as input. The output data consist of r buckets, such that at the end of the algorithm, Bucket i contains indices of all the elements in A that are equal to i.
 - Describe a decomposition based on partitioning the input data (i.e., the array A) and an appropriate mapping onto p processes. Describe briefly how the resulting parallel algorithm would work.
 - Describe a decomposition based on partitioning the output data (i.e., the set of r buckets) and an appropriate mapping onto p processes. Describe briefly how the resulting parallel algorithm would work.

- Consider seven tasks with running times of 1, 2, 3, 4, 5, 5, and 10 units, respectively. Assuming that it does not take any time to assign work to a process, compute the best- and worst-case speedup for a centralized scheme for dynamic mapping with two processes.

MODULE-2

UNIT-1

12L+8T+0P=20 Hours

INTRODUCTION TO DISTRIBUTED SYSTEMS

Goals of the Distributed Systems, Relation to parallel systems, synchronous versus asynchronous execution, design issues and challenges, Types of Distributed Systems, Distributed System Models, Hardware and software concepts related to distributed systems, middleware models.

Distributed Computing and Communication design principles: A Model of distributed executions, Models of communication networks, Global state of distributed system, Models of process communication. Communication and Coordination: Shared Memory, Consistency, Atomicity, Message- 08 Passing, Consensus, Conditional Actions, Critical Paths, Scalability, and cache coherence in multiprocessor systems, synchronization mechanism.

UNIT-2

12L+8T+0P=20 Hours

PARALLEL AND DISTRIBUTED PROGRAMMING FRAMEWORKS

Overview of CUDA, Open MP, POSIX Threads, Apache Hadoop (DFS), and current trends in parallel and distributed computing.

PRACTICES:

- Give five types of hardware resource and five types of data or software resource that can usefully be shared. Give examples of their sharing as it occurs in practice in distributed systems.
- The host computers used in peer-to-peer systems are often simply desktop computers in users' offices or homes. What are the implications of this for the availability and security of any shared data objects that they hold and to what extent can any weaknesses be overcome through the use of replication?
- Consider two communication services for use in asynchronous distributed systems. In service A, messages may be lost, duplicated or delayed and checksums apply only to headers. In service B, messages may be lost, delayed or delivered too fast for the recipient to handle them, but those that are delivered arrive with the correct contents. Describe the classes of failure exhibited by each service. Classify their failures according to their effects on the properties of validity and integrity. Can service B be described as a reliable communication service?
- Illustrate distributed design through a substantial case study, examining in detail the design of the Google infrastructure, a platform and associated middleware that supports both Google search and a set of associated web services and applications including Google Apps.
- Implementation of the parallel algorithms (on a PC-cluster under Linux platform). The programs will be based on POSIX Thread, MPI programming, Hadoop, Apache Spark etc.

SKILLS:

- ✓ Recognize parallelism in computational problems.
- ✓ Know different parallel systems and their classification.
- ✓ Design parallel algorithms for different applications.
- ✓ Compare replication schemes with respect to performance, availability and consistency concerns.
- ✓ Design, implement, and debug distributed systems.
- ✓ Implement parallel algorithms using MPI and OpenMP environments.
- ✓ Element parallel algorithms using MPI and OpenMP environments 3.0 3.5
- ✓ AVERAGE.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Apply the fundamentals of parallel and parallel computing including architectures and paradigms.	Apply	1	1
2	Analyse the various design principles of parallel algorithms.	Analyse	1	2
3	Learn the intricacies of distributed programming.	Understand	2	1
4	Develop and execute basic parallel and distributed applications using basic programming models and tools.	create	2	5

TEXT BOOKS:

1. DISTRIBUTED SYSTEMS Concepts and Design Fifth Edition. George Coulouris. Cambridge University. Jean Dollimore formerly of Queen Mary, University of London.
2. Distributed Systems Principles and Paradigms Andrew S. Tanenbaum Maarten Van Steen, 3rd Edition, 2017.

REFERENCE LINKS:

1. Introduction To Parallel Programming, Peter S. Pacheco University of San Francisco.
2. Introduction To Parallel Processing, M.Sasikumar, Dinesh Shikhare and P. Ravi Prakash, Randy Chow, T. Johnson, Distributed Operating Systems and Algorithms, Addison Wesley.
3. Ian Foster: Designing and Building Parallel Programs – Concepts and tools for Parallel Software Engineering, Pearson Publisher, 1st Edition, 2019.
4. Parallel Programming in C with MPI and Open MP Michael J.Quinn, McGrawHill Higher Education.
5. <https://hpc.llnl.gov/training/tutorials/introduction-parallel-computing-tutorial>.
6. <https://www.geeksforgeeks.org/introduction-to-parallel-computing/>.
7. <https://nptel.ac.in/>.
8. <https://www.coursera.org/>.

22CS307 SOFTWARE ENGINEERING

Hours Per Week :

L	T	P	C
2	0	2	3

PREREQUISITE KNOWLEDGE: Data base management systems, OOPs through Java.

COURSE DESCRIPTION AND OBJECTIVES:

This course focuses on the concepts of software life cycle, role of process models and methods to prepare software requirement specification document. In addition to that, it also imparts knowledge of design, development and testing of software. The objective of this course is to enable the student to develop efficient, cost effective, feasible software as per user requirements.

MODULE - 1

UNIT-1

8L+0T+8P=16 Hours

INTRODUCTION

Introduction to Software Engineering: Introduction to Software and Software engineering, Software characteristics, Software project, Software myths, Project Planning, Scheduling and Management.

Generic View of Process: Software Engineering - A layered technology, A process framework, Software Development Life Cycle (SDLC), The Capability Maturity Model Integration (CMMI).

Process Models: Conventional Model, Agile process models - Unified process model, Extreme Programming, Scrum.

UNIT-2

8L+0T+8P=16 Hours

REQUIREMENTS ENGINEERING

Requirements Engineering: Functional and Non-functional requirements, User requirements, System requirements, Requirement engineering tasks, formal requirements specification and verification, Feasibility Study.

Building the Analysis Model: Data modeling - Data objects, Attributes, Relationships, Cardinality and modality. Class based modeling - Identify analysis classes, specify attributes and Define operations.

Design Engineering: Design model, Design concepts. Creating an Architectural Design-Architectural styles and patterns.

Performing User Interface Design: Golden rules; User interface analysis and design.

PRACTICES:

Laboratory session of this course is designed in such a way that the student should complete three projects of the given type by performing the below experiments.

- Development of software requirements specification using Mind-Map tool.
- Project planning using Gantt charts.
- Project estimation using metrics.
- Capture Use Case Scenarios and model UML Use Case Diagrams.
- Model the UML state chart and Activity diagrams.
- Model the UML Class and Sequence diagrams.

MODULE - 2

UNIT-1

8L+0T+8P=16 Hours

TESTING

Testing Strategies: A strategic approach to software testing, Unit testing, Integration testing, Validation testing, System testing,

Software Engineering



Source: <https://artoftesting.com/software-engineering>

SKILLS:

- ✓ Know the software requirements and find out various ways to gather and specify them.
- ✓ Choose a process model for developing software solutions without schedule/ effort overruns and good quality.
- ✓ Analyse and model (diagrammatical representations) a software product.

Testing Tactics: Black-Box and White-Box testing techniques, Art of debugging.

Product Metrics: Metrics for analysis model; Metrics for design model, Metrics for source code; Metrics for testing; Metrics for maintenance.

UNIT-2**8L+0T+8P=16 Hours****RISK AND QUALITY MANAGEMENT**

Risk Management: Software risks, Risk identification; Risk projection; Risk refinement, Reactive vs Proactive risk strategies, RMMM.

Quality Management: Quality concepts, Formal technical reviews, Statistical Software Quality Assurance.

Computer-Aided Software Engineering (CASE): Use of appropriate CASE tools- Requirement engineering tools, Project planning tools, Testing tools.

PRACTICES:

Laboratory session of this course is designed in such a way that the student should complete three projects of the given type by performing the below experiments.

- Estimate the test coverage and Structural complexity of product using metrics.
- Develop the test cases for all the functional requirements of projects selected.
- Perform the functional testing using Selenium tool.

LIST OF PROJECTS:

Project-1: A Point-Of-Sale (PoS) System: A POS system is a computerized application used to record sales and handle payments; it is typically used in a retail store, it includes hardware components such as a computer and bar code scanner, and software to run the system. It interfaces to various service applications, such as a third-party tax calculator and inventory control. These systems must be relatively fault tolerant; that is, even if remote services are temporarily unavailable they must still be of capturing sales and handling at least cash payments. A POS system must support multiple and varied client-side terminals and interfaces such as browser, PDAs, touch-screens.

Project-2: Online Bookshop Example: Following the model of amazon.com or bn.com, design and implement an online bookstore.

Project-3: A Simulated Company: Simulate a small manufacturing company. The resulting application will enable the user to take out a loan, purchase a machine, and over a series of monthly production runs, follow the performance of their company.

Project-4: A Multi-Threaded Airport Simulation: Simulate the operations in an airport. Your application should support multiple aircrafts using several runways and gates avoiding collisions/ conflicts. Landing: an aircraft uses the runway, lands, and then taxis over to the terminal. Take-Off: an aircraft taxis to the runway and then takes off.

Project-5: An Automated Community Portal: Business in the 21st Century is above all BUSY. Distractions are everywhere. The current crop of "enterprise intranet portals" is often high noise and low value, despite the large capital expenditures it takes to stand them up. Email takes up 30 - 70% of an employee's time. Chat and Instant Messaging are either in the enterprise or just around the corner. Meanwhile, management is tasked with unforeseen and unfunded leadership and change-agent roles as well as leadership development and succession management. What is needed is a simplified, repeatable process that enhances communications within an enterprise, while allowing management and peers to self-select future leaders and easily recognize high performance team members in a dynamic way. Additionally, the system should function as a general-purpose content management, business intelligence and peer-review application. Glasscode's goal is to build that system.

Project-6: Content Management System: The goal is to enable non-technical end users to easily publish, access, and share information over the web, while giving administrators and managers complete control over the presentation, style, security, and permissions. Features: Robust Permissions System, Templates for easy custom site designs, Total control over the content, Search engine friendly URL's, Role based publishing system, Versioning control, Visitor profiling.

Project-7: An Auction Application: Several commerce models exist and are the basis for several companies like eBay.com, priceline.com etc. Design and implement an auction application that provides auctioning services. It should clearly model the various auctioneers, the bidding process, auctioning etc.

Project-8: A Notes And File Management System: During one's student years and professional career one produces a lot of personal notes, documents. All these documents are usually kept on papers or individual files on the computer. Either way the bulk of the information is often erased corrupted and eventually lost. The goal of this project is to build a distributed software application that addresses this problem. The system will provide an interface to create, organize and manage personal notes through the Internet for multiple users. The system will also allow users to collaborate by assigning permissions for multiple users to view and edit notes.

Project-9: Library Management System(LMS): The goal is to enable students and librarians to easily access and manage the library and run it smoothly. Each physical library item - book, tape cassette, CD, DVD, etc. could have its own item number. To support it, the items may be barcoded. The purpose of barcoding is to provide a unique and scannable identifier that links the barcoded physical item to the electronic record in the catalog. Barcode must be physically attached to the item, and barcode number is entered into the corresponding field in the electronic item record. Barcodes on library items could be replaced by RFID tags. The RFID tag can contain item's identifier, title, material type, etc. It is read by an RFID reader, without the need to open a book cover or CD/DVD case to scan it with barcode reader.

Project-10: Hospital Management System: Simulate to show and explain hospital structure, staff, and relationships with patients, and patient treatment terminology

Project-11: Draft Software Requirement Analysis for the following Problem Statement: Fuel Delivery System: An unattended petrol (gas) pump system that includes a credit card reader. The customer swipes the card through the reader and then specifies the amount of fuel required. The fuel is delivered, and the customer's account debited.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Use basic concepts of software engineering for designing software product.	Usage	1	1, 11
2	Compare different process models and identify appropriate process model based on project requirements.	Evaluation	1	2, 4
3	Build Software Requirement Specification (SRS) document for any software product.	Design	1	3, 5
4	Design of solutions using UML diagrams like Use case, Sequence diagrams etc.	Design	1	3, 4, 5
5	Create an appropriate architecture for a given project that meets all quality constraints.	Create	2	5
6	Apply different testing techniques to ensure bug free software and metrics to measure the software size, complexity, and budget etc.	Apply	2	4, 5, 11

TEXT BOOKS:

1. Roger S. Pressman, "Software Engineering, A practitioner's Approach", 6th Edition, McGrawHill International Edition, 2008.
2. Booch G., Rumbaugh J. and Jacobsons I, "The Unified Modeling Language User Guide", 2nd Edition, Addison Wesley, 2005.

REFERENCE BOOKS:

1. Simon Sennet, Steve McRobb and Ray Farmer, "Object Oriented Systems Analysis and Design, 2nd edition, 2004.
2. Dr. Pankaj Jalote "Software Engineering: A Precise Approach"-edition 2010.

22CS808 MACHINE LEARNING

Hours Per Week :

L	T	P	C
3	0	2	4

PREREQUISITE KNOWLEDGE: Probability and Linear algebra, Python programming.

COURSE DESCRIPTION AND OBJECTIVES:

This course provides a broad introduction to various machine learning concepts including Supervised learning (parametric/non-parametric algorithms, support vector machines, kernels, neural networks) and Unsupervised learning (clustering, dimensionality reduction) methods. Students will get an understanding of various challenges of Machine Learning and will be able to decide on model complexity. Numerous case studies introduced in this course allow the students to apply machine-learning algorithms in computer vision, medical imaging, audio, and text domains. Laboratory experiments of this course will introduce students to advanced Machine Learning Python libraries such as Scikit-Learn, Matplotlib, and many other recent ML-related APIs. The course is designed such that the students get enough hands-on experience with a major focus on the practical implementation of theoretical concepts.

MODULE-1

UNIT-1

14L+0T+8P=22 Hours

INTRODUCTION

What is machine learning? Machine learning applications; Types of Learning: Supervised learning; Un-supervised learning; Reinforcement learning, Applications of Probability and statistics.

Model Training Essentials: Re-sampling methods: Bias–Variance Trade-off. Hypothesis Testing and Variable Selection, Sub sampling and Upsampling, SMOTE; Cross Validation (validation set, Leave-One-Cut (LOO), k-fold strategies) and bootstrap; Evaluation measures-Error functions, Confusion Matrix, Accuracy, Precision and Recall, F1 Score.

Regression Analysis: Linear Regression, Simple and Multiple Linear Regression, Polynomial Regression, Logistic Regression, Multi nominal Regression. Ordinary Least Squares Method, Model Shrinkage-Ridge, and LASSO regression.

UNIT-2

10L+0T+8P=18 Hours

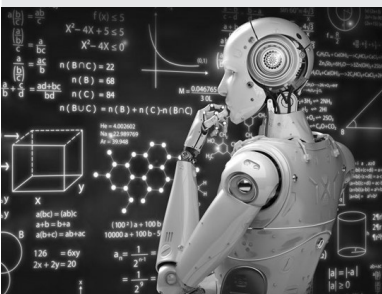
FEATURE SELECTION

Feature Selection Strategies: Problem statement and Uses, Filter methods, Wrapper methods, Embedded methods. Branch and bound algorithm, Sequential forward/backward selection algorithms.

Dimensionality Reduction: Singular value decomposition, matrix factorization, Linear discriminant analysis, Principal components analysis.

PRACTICES:

- Apply the following tasks to any given dataset:
 - a) Load and visualize data.
 - b) Check out and replace missing values.
 - c) Encode the Categorical data.
 - d) Splitting the dataset into Training and Test set.
 - e) Splitting the dataset into k-folds.
 - f) Feature scaling.



Source: <https://www.forbes.com/sites/kalevleetaru/2019/01/15/why-machine-learning-needs-semantics-not-just-statistics/>

- House price prediction:
 - a) Create a model that predicts a continuous value (price) from input features (square footage, number of bedrooms and bathrooms).
 - b) Implement a univariate Model using Least Squares and plot best-fit line.
 - c) Implement a multivariate Model using Least Squares and plot best-fit line.
 - d) Retrieve model error and model coefficients.
 - e) Observe Variance Inflation Factor (VIF).
 - f) Implement Ridge regression model.
 - g) Implement LASSO regression model.
 - h) Report your observations on the above models for house prediction.
- Heart disease prediction:
 - a) Implement a logistic regression model to predict whether an individual is suffering from heart disease or not.
 - b) Evaluate and compare model performance using the following validation approaches:
 - i. Validation set approach.
 - ii. K-fold cross validation.
 - iii. Stratified K-fold cross validation.
 - iv. LOO strategy.
 - c) Plot Confusion matrix.
 - d) Report performance of the model in terms of the following metrics:
 - i. Accuracy.
 - ii. Precision-Recall.
 - iii. F1 Score.
 - e) Report your observations and explain when to use what type of measures.
- Implement the Polynomial Regression algorithm to fit data points. Select the appropriate data set for your experiment and draw graphs.
- Working with imbalanced datasets:
 - a) Load an imbalanced dataset and visualize imbalance in the data as a bar plot.
 - b) Implement KNN model for classification.
 - c) Balance the dataset using:
 - i. Random Over sampling.
 - ii. Random Under sampling.
 - iii. SMOTE.
 - d) Implement KNN model for classifying data balanced in the above steps.
 - e) Report your observations on the performance of models trained using balanced and imbalanced data.
- Perform effective feature selection in a given dataset using any one of the feature selection techniques.
- Dimension Reduction:
 - a) Load a dataset and Implement Bayes classification model.
 - b) Apply dimension reduction using:
 - i. Principal Component Analysis
 - ii. Linear Discriminant Analysis
 - c) Apply the model on data with reduced dimension.
 - d) Compare and contrast model performance in each case.

SKILLS:

- ✓ Statistical data analysis.
- ✓ Classify / Cluster data.
- ✓ Tool usage for developing ML applications.

MODULE-2**UNIT-1****16L+0T+8P=24 Hours****CLASSIFICATION**

Classification: Binary, Multi-class and Multi-label Classification; K-Nearest Neighbours, Support Vector Machines, Decision Trees, The Naïve Bayes' Classifier, Class Imbalance, Perceptron ANN model.

Ensemble Methods: Ensemble Learning Model Combination Schemes, Voting, Error-Correcting Output Codes, Bagging: Random Forest Trees, Boosting: Adaboost, Stacking.

UNIT-2**8L+0T+8P=16 Hours****CLUSTERING**

Clustering: Different distance functions and similarity measures, K-means clustering, Medoids, Hierarchical Clustering-Single linkage and Complete linkage clustering, Graph based Clustering -MST, DBSCAN, Spectral clustering.

PRACTICES:

- Implement and demonstrate the FIND-S algorithm for finding the most specific hypothesis based on a given set of training data samples. Read the training data from a .CSV file.
- Implement the naïve Bayesian classifier for a sample training data set stored as a.csv file. Compute the accuracy of the classifier, considering few test data sets.
- Assuming a set of spam or not-spam mails that need to be classified, use the naïve Bayesian classifier model to perform this task. Calculate the accuracy, precision, and recall for your data set.
- Implement k-Nearest Neighbor algorithm to classify the iris data set. Print both correct and wrong predictions. Python ML library classes can be used for this problem.
- Demonstrate the working of the decision tree-based ID3 algorithm. Use an appropriate data set for building the decision tree and apply this knowledge to classify a new sample?
- Build a model using SVM with different kernels.
- Implement and build models using the following Ensemble techniques.
 - a) Bagging.
 - b) Boosting: Adaboost, Stacking.
- Build a model to perform Clustering using K-means after applying PCA and determining the value of K using the Elbow method.
- Unsupervised Modeling:
 - a) Cluster the data using the following models:
 - i. Spectral Clustering.
 - ii. K-medoids.
 - iii. DBSCAN.
 - iv. Hierarchical Clustering.
 - b) Compare and contrast model performance in each case.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Apply a wide variety of learning algorithms such as Probabilistic, Discriminative and Generative algorithms for a given application.	Apply	1, 2	1
2	Design an end-to-end Machine-learning model to realize solutions for real-world problems.	Apply	1	3
3	Implement various machine learning models using advanced ML tools.	Create	1, 2	5
4	Analyze and evaluate the performance of various machine learning models approaches on different kinds of data.	Analyze	2	2

TEXT BOOKS:

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